Year 7 Set 3 · White Rose · Lesson 1

Form and solve one-step equations

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Today I will learn to...

Goals

1. Solve one-step equations

2. Form and solve equations from words

Do now: retrieval practice

4 mins

1. Simplify a + a + a

   3a

2. Find 3x when x = 7

   21

3. Work out −4 + 9

   5

4. What goes in the box? ☐ + 8 = 15

   7

Objective 1

Solve one-step equations

Key idea

Learn

!

An equation is balanced: both sides are equal. To solve it, do the inverse operation to both sides. + and − undo each other; × and ÷ undo each other.

Objective 1: Solve one-step equations

I do: example 1

I do

Solve x + 7 = 19

1. The inverse of + 7 is − 7
2. x + 7 − 7 = 19 − 7
3. x = 12
4. Check: 12 + 7 = 19 ✓

I do: example 2

I do

Solve $y - 4.5 = 10$

1. The inverse of -4.5 is $+4.5$
2. $y = 10 + 4.5$
3. $y = 14.5$
4. Check: $14.5 - 4.5 = 10$ ✓

I do: example 3

I do

Solve 6m = 42 and n/5 = 3

1. 6m means 6 × m, so divide both sides by 6: m = 7
2. n/5 means n ÷ 5, so multiply both sides by 5
3. n = 15
4. Check: 6 × 7 = 42 and 15 ÷ 5 = 3 ✓

Mini whiteboards

We do

1. Solve $a - 9 = 14$

Answer: $a = 23$

2. Solve $4b = 36$

Answer: $b = 9$

3. Solve $c + 17 = 40$

Answer: $c = 23$

4. Solve $d/5 = 8$

Answer: $d = 40$

You do: choose your challenge

You do

Bronze

Solve $x + 15 = 32$

Silver

Solve $7y = 63$

Gold

Solve $t/3 = 12$ and $20 = k - 6$

Platinum

Solve $0.4w = 2$

You do: answers

Check

Bronze

x = 17

Silver

y = 9

Gold

t = 36, k = 26

Platinum

w = 2 ÷ 0.4 = 5

If you finish early

Extra

Bronze

Solve $x - 6 = 13$

$x = 19$

Silver

Solve $9y = 72$

$y = 8$

Gold

Solve $p/4 = 3.5$

$p = 14$

Platinum

Solve $2.5m = 10$ and $n - 4.7 = 3.3$

$m = 4, n = 8$

Exam-style question 1

1 mark

Solve $8x = 56$

Space for working

Mark scheme: x = 7 (B1)

Exam-style question 2

2 marks

Solve $p/4 = 2.5$

Space for working

Mark scheme: $p = 2.5 \times 4$ (M1) $= 10$ (A1)

Objective 1 achieved

Solve one-step equations



Objective 2

Form and solve equations from words

✓ Achieved: Solve one-step equations

Key idea

Learn

!

Choose a letter for the unknown. Turn the words into an equation, solve it, then answer the question in words.

Objective 2: Form and solve equations from words

I do: example 1

I do

I think of a number and add 12. The answer is 30. What is my number?

1. Let the number be n
2. n + 12 = 30
3. n = 30 − 12 = 18
4. My number is 18

I do: example 2

I do

A pack of 6 pens costs £4.50. Find the cost of one pen.

1. Let one pen cost £p
2. 6p = 4.50
3. p = 4.50 ÷ 6 = 0.75
4. One pen costs 75p

I do: example 3

I do

The perimeter of a square is 36 cm. Find the length of one side.

1. Let one side be s cm
2. $4s = 36$
3. $s = 36 \div 4 = 9$
4. Each side is 9 cm

Mini whiteboards

We do

1. I think of a number and subtract 8. The answer is 15. Find the number.

Answer: 23

2. 5 equal pieces of ribbon make 3.5 m. How long is each piece?

Answer: 0.7 m (70 cm)

3. I think of a number and multiply it by 6. The answer is 54. Find the number.

Answer: 9

4. A pack of 4 drinks costs £3. Find the cost of one drink.

Answer: 75p

You do: choose your challenge

You do

Bronze

I think of a number and add 9. The answer is 25. Find the number.

Silver

A number multiplied by 8 is 72. Find the number.

Gold

Ben gives away 14 sweets and has 29 left. Form and solve an equation.

Platinum

A rectangle has length 3w and width w. Its perimeter is 48 cm. Find w.

You do: answers

Check

Bronze

16

Silver

9

Gold

$s - 14 = 29$, so $s = 43$

Platinum

$3w + w + 3w + w = 8w = 48$, so $w = 6$

If you finish early

Extra

Bronze

I think of a number and add 15. The answer is 40. Find the number.

25

Silver

I think of a number and divide it by 3. The answer is 9. Find the number.

27

Gold

An equilateral triangle has a perimeter of 22.5 cm. Form and solve an equation to find one side.

3s = 22.5, s = 7.5 cm

Platinum

Ella is x years old. Her gran is 6 times as old as Ella. Her gran is 72. Form and solve an equation.

6x = 72, x = 12

Exam-style question 1

2 marks

Sam is x years old. His mum is 4 times as old as Sam. His mum is 44. Form an equation and find x.

Space for working

Mark scheme: 4x = 44 (M1). x = 11 (A1)

Exam-style question 2

2 marks

An equilateral triangle has a perimeter of 27.6 cm. Form an equation and find the length of one side.

Space for working

Mark scheme: 3s = 27.6 (M1). s = 9.2 cm (A1)

Objective 2 achieved

Form and solve equations from words



Real life

Apply

A taxi charges £2 per mile. A journey costs £17. Form and solve an equation to find how many miles it was.

2m = 17, so m = 8.5 miles

Extension: stretch yourself

Stretch

1. Solve 2x + 5 = 21

2x = 16, so x = 8

2. Two angles on a straight line are x and 3x. Form and solve an equation.

x + 3x = 180, so 4x = 180 and x = 45°

Exit ticket

Exit

1. Solve $x - 17 = 25$ **(1 mark)**

2. Solve $9y = 108$ **(1 mark)**

3. I think of a number and divide it by 4. The answer is 7.5. What was my number? **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 x = 42 [1]

Q2 y = 12 [1]

Q3 n/4 = 7.5 (M1). n = 30 (A1) [2]

Total: _____ / 4

Today I achieved

- ✔ **Solve one-step equations**
- ✔ **Form and solve equations from words**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework

Due next lesson

1. Solve $m + 23 = 51$ [1]

2. Solve $12k = 84$ [1]

3. Solve $n/6 = 9$ [1]

4. Solve $x - 3.8 = 6.2$ [1]

5. A box of 8 identical books weighs 6.4 kg. Form and solve an equation to find the weight of one book. [2]

Extra practice: Corbettmaths → search "Solving one-step equations"